ANA500 Micro-Project 1

Commented Python code and executed results
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Source: supplied airline.csv. Run the notebook cells in order. The code preserves raw data and explains cleaning decisions. Repository: https://github.com/ChalesManyang/ANA500-Micro-Projects.

Acquire: load the supplied file and inspect its structure.

```
from pathlib import Path
import numpy as np
import pandas as pd
# Keep airline.csv beside this notebook, or edit DATA_PATH below.
DATA_PATH = Path('airline.csv')
if not DATA_PATH.exists():
    DATA_PATH = Path('../airline.csv')
if not DATA_PATH.exists():
    DATA_PATH = Path('../../upload/airline.csv')
raw = pd.read_csv(DATA_PATH)
print('Raw rows and columns:', raw.shape)
print(raw.dtypes.to_string())
print('Missing values:', raw.isna().sum()[raw.isna().sum() > 0].to_dict())
```

Executed output

```
Raw rows and columns: (129880, 25)
Unnamed: 0                              int64
id                                      int64
Gender                                 object
Customer Type                          object
Age                                     int64
Type of Travel                         object
Class                                  object
Flight Distance                         int64
Inflight wifi service                   int64
Departure/Arrival time convenient       int64
Ease of Online booking                  int64
Gate location                           int64
Food and drink                          int64
Online boarding                         int64
Seat comfort                            int64
Inflight entertainment                  int64
On-board service                        int64
Leg room service                        int64
Baggage handling                        int64
Checkin service                         int64
Inflight service                        int64
Cleanliness                             int64
Departure Delay in Minutes              int64
Arrival Delay in Minutes              float64
satisfaction                           object
Missing values: {'Arrival Delay in Minutes': 393}
```

Prepare: preserve raw data and create an independent working copy.

```
df = raw.copy()
for col in df.select_dtypes(include='object'):
    df[col] = df[col].str.strip()
# The exported row counter is not a passenger attribute.
df = df.drop(columns=[c for c in df if c.startswith('Unnamed:')])
# Check complete duplicate records before considering any removal.
print('Exact duplicate rows:', int(df.duplicated().sum()))
print('Repeated passenger IDs:', int(df['id'].duplicated().sum()))
# Preserve valid zero ratings; their meaning is undocumented in this upload.
# Never assume zero means missing without a data dictionary.
ratings = list(df.loc[:, 'Inflight wifi service':'Cleanliness'].columns)
```

```
assert df[ratings].apply(lambda s: s.between(0, 5).all()).all()
assert df['Age'].between(0, 120).all()
assert (df[['Flight Distance', 'Departure Delay in Minutes']].to_numpy() >= 0).all()
print('Rating range:', int(df[ratings].min().min()), int(df[ratings].max().max()))
# Keep missing arrival delays as NaN. Project 4 excludes them as unknown targets.
print(df[['Age','Flight Distance','Departure Delay in Minutes',
          'Arrival Delay in Minutes']].describe().round(2).to_string())
print('Satisfied share (%):', round(df['satisfaction'].eq('satisfied').mean()*100, 2))
# NumPy arrays support explicit missing-value-aware numerical summaries.
arrivals = df['Arrival Delay in Minutes'].to_numpy(dtype=float)
print('Arrival mean / median:', np.nanmean(arrivals), np.nanmedian(arrivals))
print('Known arrival values:', int(np.isfinite(arrivals).sum()))
print('Preparation complete. No missing targets were filled.')
```

Executed output

```
Exact duplicate rows: 0
Repeated passenger IDs: 0
Rating range: 0 5
             Age  Flight Distance  Departure Delay in Minutes  Arrival Delay in Minutes
count  129880.00        129880.00                   129880.00                 129487.00
mean       39.43          1190.32                       14.71                     15.09
std        15.12           997.45                       38.07                     38.47
min         7.00            31.00                        0.00                      0.00
25%        27.00           414.00                        0.00                      0.00
50%        40.00           844.00                        0.00                      0.00
75%        51.00          1744.00                       12.00                     13.00
max        85.00          4983.00                     1592.00                   1584.00
Satisfied share (%): 43.45
Arrival mean / median: 15.09112883918849 0.0
Known arrival values: 129487
Preparation complete. No missing targets were filled.
```